

Naimark's problem

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MATHLIBANNEX / PROGRESSIVE RESEARCH COMPANION

The mathematical goal

A single unital, simple, infinite-dimensional C^* -algebra has one unitary-equivalence class of nonzero irreducible representations and a faithful tracial representation on a separable Hilbert space. The normalized trace of its CAR subalgebra extends uniquely among all states of the same generated algebra.

Exact source: MathlibAnnex v0.4.0 Project entry · Source manifest

Scope

The principal construction concerns complex C^* -algebras and nonzero irreducible representations. Its fixed algebra and separable faithful tracial representation require no CH hypothesis. CH enters only in the statement equating the continuum with aleph one and the corresponding density-character existence result. The general reverse density obstruction also allows nonunital algebras.

68 selected Declaration Cards form this Companion, with 1,044 dependency relations, 122 displayed edges and levels 0–15. The CH consequence is a separate route reusing two density Cards.

Mathematical routes

CAR foundations and canonical states

12 Cards · levels 0–4

Finite-stage purification and local transport

6 Cards · levels 0–2

Pure-state homogeneity and shell matching

8 Cards · levels 0–3

Direct sums of GNS representations and shell unitaries

7 Cards · levels 0–3

Fixed subspaces and reconstruction of the generators

8 Cards · levels 0–2

Unitary equivalence of irreducible representations

9 Cards · levels 0–4

Trace extensions and faithful representations on a separable space

9 Cards · levels 0–3

One C^* -algebra and its representation-theoretic properties

7 Cards · levels 0–3

Density and cardinality

2 Cards · levels 0–1

CH and the existence of a counterexample of norm density aleph one

3 Cards · levels 0–2

Boundary Inputs

The displayed edges preserve dependency paths through omitted helpers. Levels count selected predecessors within this scope. Mathematical citations remain distinct from formal dependencies.

Exact source and provider boundary · Earlier 466-declaration Project view and PDF

Dependency-first reading route

Levels belong to this reading scope. Follow prerequisites or uses to focus the route.

Level 0

LEVEL 0

The normalized trace on a finite CAR stage

`MathlibAnnex.CStarAlgebra.CAR.stageTrace`

The normalized matrix trace provides a positive tracial functional compatible with the CAR inclusions.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this scope

USED BY IN THIS PROJECT

Constructing the normalized trace on the completed CAR algebra

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LEVEL 0

The CAR algebra as a completion of finite matrix stages

`MathlibAnnex.CStarAlgebra.CAR.Limit`

Names the completed CAR algebra in which finite matrix calculations can be extended by norm approximation.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this scope

USED BY IN THIS PROJECT

The product-vector state on the completed CAR algebra, Approximately inner homogeneity of pure CAR states, A finite-row average centralizes its matrix stage, The decreasing root projections in the CAR completion, Constructing the normalized trace on the completed CAR algebra, The completed CAR algebra is infinite-dimensional, A two-sided inner intertwining sequence for two states

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LEVEL 0

Exact projection links for one approximately inner automorphism

`MathlibAnnex.CStarAlgebra.exists_shell_family_of_approximately_inner`

A point-norm approximation followed by close-projection conjugacy gives exact support identities for an arbitrary projection family.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this scope

USED BY IN THIS PROJECT

Choosing the CAR shell family from proved homogeneity

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LEVEL 0

A fixed representative family of CAR shell data

`MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellFamily`

One family records a transporting automorphism and exact shell links for each selected pure-state class, with an explicitly fixed root component.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this scope

USED BY IN THIS PROJECT

Choosing the CAR shell family from proved homogeneity, The matching GNS fiber retains exactly its cyclic line, Shell matching fixes the trace of transported flags, An inequivalent GNS fiber has no residual common range, Reconstructing a represented unitary from its shells and residual corner, Assembling selected GNS cyclic subspaces into an isometric source representation

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LEVEL 0

The selected GNS representation of a pure-state class

`MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation`

A choice of representative pure states, fixed literally at a root, gives one concrete GNS representation per equivalence class.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this scope

USED BY IN THIS PROJECT

The matching GNS fiber retains exactly its cyclic line, Distinct GNS classes have no unitary intertwiner, The atomic direct sum of the selected CAR representations

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LEVEL 0

The closed operator algebra generated by a representation and extra operators

`MathlibAnnex.CStarAlgebra.AtomicConstruction.concreteTarget`

Places a represented algebra and a chosen family of bounded operators in one concrete closed algebra.

IMMEDIATE PREREQUISITES IN THIS PROJECT

None in this scope

USED BY IN THIS PROJECT

An irreducible operator algebra constructed from projection shells, Reconstructing a represented unitary from its shells and residual corner

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Level 1

LEVEL 1

The product-vector state on the completed CAR algebra

`MathlibAnnex.CStarAlgebra.CAR.rootState`

Compatible evaluation at the distinguished matrix coordinate extends continuously to the CAR completion.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The CAR algebra as a completion of finite matrix stages

USED BY IN THIS PROJECT

Root compression becomes scalar in norm, Purity of the completed root state

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LEVEL 1

The completed CAR algebra is infinite-dimensional

`MathlibAnnex.CStarAlgebra.CAR.not_finiteDimensional`

Rules out finite dimension by retaining matrix subalgebras of unbounded dimension.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The CAR algebra as a completion of finite matrix stages

USED BY IN THIS PROJECT

The shell-generated algebra is not an algebra of all compact operators, An irreducible CAR representation has no nonzero compact image

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LEVEL 1

A finite-row average centralizes its matrix stage

`MathlibAnnex.CStarAlgebra.CAR.ofStage_commute_rowAverage`

Turns an arbitrary element of the completed CAR algebra into one commuting with a prescribed finite matrix stage.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The CAR algebra as a completion of finite matrix stages

USED BY IN THIS PROJECT

Exact vector transport along an almost central unitary path, Approximating another vector state along a unitary path

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LEVEL 1

Constructing the normalized trace on the completed CAR algebra

`MathlibAnnex.CStarAlgebra.CAR.trace`

Carries compatible bounded matrix traces through the algebraic limit, normed union and completion.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The normalized trace on a finite CAR stage, The CAR algebra as a completion of finite matrix stages

USED BY IN THIS PROJECT

Shell matching fixes the trace of transported flags, Extending the CAR trace to the fixed shell target, Normalization and the trace identity determine the CAR trace

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LEVEL 1

The decreasing root projections in the CAR completion

`MathlibAnnex.CStarAlgebra.CAR.rootFlag`

The distinguished stage corners give a nested sequence of projections supporting the root state.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The CAR algebra as a completion of finite matrix stages

USED BY IN THIS PROJECT

Root compression becomes scalar in norm, Shell matching fixes the trace of transported flags, Reconstructing a represented unitary from its shells and residual corner

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LEVEL 1

Approximately inner homogeneity of pure CAR states

`MathlibAnnex.CStarAlgebra.CAR.PureStateHomogeneity`

The homogeneity property asks for exact state transport by an automorphism that is locally approximable by inner automorphisms.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The CAR algebra as a completion of finite matrix stages

USED BY IN THIS PROJECT

Two-sided inner sequences imply pure-state homogeneity

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LEVEL 1

A two-sided inner intertwining sequence for two states

`MathlibAnnex.CStarAlgebra.CAR.HasInnerIntertwiningSequence`

An inner sequence records both forward and actual inverse convergence data before any limiting automorphism is constructed.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The CAR algebra as a completion of finite matrix stages

USED BY IN THIS PROJECT

Two-sided inner sequences imply pure-state homogeneity, Pure CAR states admit two-sided inner intertwining sequences

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LEVEL 1

Distinct GNS classes have no unitary intertwiner

`MathlibAnnex.CStarAlgebra.PureState.no_unitaryIntertwiner_selectedRepresentation`

The class index makes the selected pure-GNS family pairwise unitarily inequivalent.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The selected GNS representation of a pure-state class

USED BY IN THIS PROJECT

An inequivalent GNS fiber has no residual common range, The atomic-shell construction on selected pure-GNS fibers, Assembling selected GNS cyclic subspaces into an isometric source representation

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LEVEL 1

An irreducible operator algebra constructed from projection shells

`MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_atomicShellModel`

Shell partial isometries with rank-one limiting defects can be completed to unitary links that join inequivalent irreducible fibers.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The closed operator algebra generated by a representation and extra operators

USED BY IN THIS PROJECT

The atomic-shell construction on selected pure-GNS fibers

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Level 2

LEVEL 2

Purity of the completed root state

`MathlibAnnex.CStarAlgebra.CAR.isPureState_rootState`

Passes extremality of the root vector states from every finite matrix stage to the completed CAR algebra.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The product-vector state on the completed CAR algebra

USED BY IN THIS PROJECT

Choosing the CAR shell family from proved homogeneity, The matching GNS fiber retains exactly its cyclic line, The atomic direct sum of the selected CAR representations, An inequivalent GNS fiber has no residual common range

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LEVEL 2

Normalization and the trace identity determine the CAR trace

`MathlibAnnex.CStarAlgebra.CAR.eq_trace_of_apply_one_of_mul_comm`

Proves uniqueness without assuming positivity of the competing continuous functional.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Constructing the normalized trace on the completed CAR algebra

USED BY IN THIS PROJECT

The fixed shell target has a unique tracial state

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LEVEL 2

Root compression becomes scalar in norm

`MathlibAnnex.CStarAlgebra.CAR.tendsto_norm_compressionError`

Extends an exact rank-one compression identity on finite stages to a pointwise norm limit on the completed algebra.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The product-vector state on the completed CAR algebra, The decreasing root projections in the CAR completion

USED BY IN THIS PROJECT

The matching GNS fiber retains exactly its cyclic line, An inequivalent GNS fiber has no residual common range, Simplicity of the completed CAR algebra, A common root vector realizes every selected state through the generators, Assembling selected GNS cyclic subspaces into an isometric source representation

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LEVEL 2

Two-sided inner sequences imply pure-state homogeneity

`MathlibAnnex.CStarAlgebra.CAR.homogeneity_of_innerIntertwiningSequences`

The forward and inverse limits close the homogeneity argument once an intertwining sequence is supplied for every pure-state pair.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Approximately inner homogeneity of pure CAR states, A two-sided inner intertwining sequence for two states

USED BY IN THIS PROJECT

Pure-state homogeneity of the completed CAR algebra

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LEVEL 2

The atomic-shell construction on selected pure-GNS fibers

`MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_pureAtomicShellModel`

Pure-state GNS data supplies the irreducible and inequivalent fibers required by the generic shell construction.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Distinct GNS classes have no unitary intertwiner, An irreducible operator algebra constructed from projection shells

USED BY IN THIS PROJECT

Realizing a shell family by a faithful irreducible operator algebra

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LEVEL 2

Shell matching fixes the trace of transported flags

`MathlibAnnex.CStarAlgebra.CAR.trace_transportedFlag`

Recovers exact trace values from the initial and final supports of shell links.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The decreasing root projections in the CAR completion, Constructing the normalized trace on the completed CAR algebra, A fixed representative family of CAR shell data

USED BY IN THIS PROJECT

Transported flags vanish on vectors generated by a trace vector

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Level 3

LEVEL 3

Simplicity of the completed CAR algebra

`MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_limit`

Every nonzero closed two-sided ideal of the CAR algebra contains the identity.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Root compression becomes scalar in norm

USED BY IN THIS PROJECT

An irreducible CAR representation has no nonzero compact image

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LEVEL 3

The atomic direct sum of the selected CAR representations

`MathlibAnnex.CStarAlgebra.CAR.selectedAtomicRepresentation`

All selected pure-GNS fibers of the completed CAR algebra act together on one Hilbert direct sum.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The selected GNS representation of a pure-state class, Purity of the completed root state

USED BY IN THIS PROJECT

Realizing a shell family by a faithful irreducible operator algebra, Reconstructing a represented unitary from its shells and residual corner, Assembling selected GNS cyclic subspaces into an isometric source representation

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LEVEL 3

Transported flags vanish on vectors generated by a trace vector

`MathlibAnnex.CStarAlgebra.CAR.tendsto_transportedFlag_orbit_zero`

Turns decay of projection traces into norm convergence on each source-orbit vector.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Shell matching fixes the trace of transported flags

USED BY IN THIS PROJECT

The target represented on the CAR trace GNS space, Pointed unitary transport between trace-cyclic target representations

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LEVEL 3

The matching GNS fiber retains exactly its cyclic line

`MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_rankOne`

Identifies the residual projection of a transported CAR flag in its matching pure-state representation.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Root compression becomes scalar in norm, The selected GNS representation of a pure-state class, Purity of the completed root state, A fixed representative family of CAR shell data

USED BY IN THIS PROJECT

The atomic common range is one embedded GNS line

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LEVEL 3

An inequivalent GNS fiber has no residual common range

`MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_zero`

Uses compression and cyclic transport to exclude fixed vectors in every other chosen pure-state class.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Root compression becomes scalar in norm, Distinct GNS classes have no unitary intertwiner, Purity of the completed root state, A fixed representative family of CAR shell data

USED BY IN THIS PROJECT

The atomic common range is one embedded GNS line

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Level 4

LEVEL 4

An irreducible CAR representation has no nonzero compact image

`MathlibAnnex.CStarAlgebra.CAR.eq_zero_of_isCompactOperator_image`

Simplicity and infinite dimensionality rule out compact operators coming from nonzero CAR elements.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Simplicity of the completed CAR algebra, The completed CAR algebra is infinite-dimensional

USED BY IN THIS PROJECT

Lifting a corner involution to an ambient CAR unitary, Realizing a finite Gram matrix in a represented CAR corner

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LEVEL 4

The atomic common range is one embedded GNS line

`MathlibAnnex.CStarAlgebra.CAR.iInf_range_atomic_transportFlag_eq_span`

Assembles the matching and inequivalent fiber calculations in an arbitrary Hilbert direct sum.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The matching GNS fiber retains exactly its cyclic line, An inequivalent GNS fiber has no residual common range

USED BY IN THIS PROJECT

Every irreducible representation is unitarily equivalent to the inclusion, Realizing a shell family by a faithful irreducible operator algebra

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LEVEL 4

Reconstructing a represented unitary from its shells and residual corner

MathlibAnnex.CStarAlgebra.CAR.exists_targetShellReconstruction

Separates a represented generator into a strong shell sum and the exact operator between its limiting fixed spaces.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The atomic direct sum of the selected CAR representations, The decreasing root projections in the CAR completion, A fixed representative family of CAR shell data, The closed operator algebra generated by a representation and extra operators

USED BY IN THIS PROJECT

The target represented on the CAR trace GNS space, Pointed unitary transport between trace-cyclic target representations, An irreducible target representation has a surviving fixed space

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LEVEL 4

Assembling selected GNS cyclic subspaces into an isometric source representation

MathlibAnnex.CStarAlgebra.CAR.exists_selectedAtomicCyclicIsometry

Joins mutually orthogonal pure-state cyclic copies and controls the limiting flag projections on their sum.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Root compression becomes scalar in norm, Distinct GNS classes have no unitary intertwiner, The atomic direct sum of the selected CAR representations, A fixed representative family of CAR shell data

USED BY IN THIS PROJECT

The cyclic sum fills every irreducible target representation

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Level 5

LEVEL 5

Realizing a finite Gram matrix in a represented CAR corner

`MathlibAnnex.CStarAlgebra.CAR.exists_rootCornerFamily_with_gram`

The infinite-dimensional root corner contains an isometric copy of any finite family of vectors.

IMMEDIATE PREREQUISITES IN THIS PROJECT

An irreducible CAR representation has no nonzero compact image

USED BY IN THIS PROJECT

Purifying a vector state on a finite CAR stage

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LEVEL 5

Lifting a corner involution to an ambient CAR unitary

`MathlibAnnex.CStarAlgebra.CAR.exists_liftedCornerExponential_apply_eq_involution`

A prescribed unitary involution on a represented root corner can be realized on finitely many vectors by one lifted corner exponential.

IMMEDIATE PREREQUISITES IN THIS PROJECT

An irreducible CAR representation has no nonzero compact image

USED BY IN THIS PROJECT

Exact vector transport along an almost central unitary path, Approximating another vector state along a unitary path

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LEVEL 5

Realizing a shell family by a faithful irreducible operator algebra

`MathlibAnnex.CStarAlgebra.CAR.exists_completedAtomicShellModel`

Constructs unitary links between pure-state summands and an irreducible algebra containing a faithful copy of CAR.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The atomic common range is one embedded GNS line, The atomic direct sum of the selected CAR representations, The atomic-shell construction on selected pure-GNS fibers

USED BY IN THIS PROJECT

The ambient inclusion of the fixed shell-family target, The CAR source map into the fixed shell-family target, The C*-algebra generated by the CAR representation and the shell unitaries

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LEVEL 5

An irreducible target representation has a surviving fixed space

`MathlibAnnex.CStarAlgebra.CAR.exists_nonzero_targetFixedSpace`

Rules out simultaneous disappearance of all limiting CAR flags by passing reduction through the reconstructed generators.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Reconstructing a represented unitary from its shells and residual corner

USED BY IN THIS PROJECT

A common root vector realizes every selected state through the generators

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Level 6

LEVEL 6

Purifying a vector state on a finite CAR stage

`MathlibAnnex.CStarAlgebra.CAR.exists_unitVector_vectorFunctional_eq_on_stage`

Every finite-stage restriction of a unit vector state can be realized by a unit vector in any irreducible CAR representation.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Realizing a finite Gram matrix in a represented CAR corner

USED BY IN THIS PROJECT

Approximating another vector state along a unitary path, Cross-representation state approximation with a protected finite set

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LEVEL 6

Exact vector transport along an almost central unitary path

`MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_exact_unitary_path_apply_eq_and_conjugate_sub_norm_lt`

Finite matrix-state tests ensure exact transport while protecting a prescribed finite set throughout the path.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Lifting a corner involution to an ambient CAR unitary, A finite-row average centralizes its matrix stage

USED BY IN THIS PROJECT

Cross-representation state approximation with a protected finite set

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LEVEL 6

A common root vector realizes every selected state through the generators

`MathlibAnnex.CStarAlgebra.CAR.exists_targetDefectVectorFamily`

Builds a compatible family of unit vectors from a surviving residual space and identifies their CAR vector states.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Root compression becomes scalar in norm, An irreducible target representation has a surviving fixed space

USED BY IN THIS PROJECT

The cyclic sum fills every irreducible target representation

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LEVEL 6

The CAR source map into the fixed shell-family target

`MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom`

Restricting the codomain of the selected atomic representation gives the source map into one fixed generated target.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Realizing a shell family by a faithful irreducible operator algebra

USED BY IN THIS PROJECT

The shell-generated algebra is not an algebra of all compact operators, Closed-ideal simplicity of the shell-generated algebra, Pointed unitary transport between trace-cyclic target representations, Extending the CAR trace to the fixed shell target

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LEVEL 6

The ambient inclusion of the fixed shell-family target

`MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion`

The same concrete target has a literal representation on the selected atomic Hilbert space.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Realizing a shell family by a faithful irreducible operator algebra

USED BY IN THIS PROJECT

The ambient representation of the fixed algebra, Closed-ideal simplicity of the shell-generated algebra, Unitary equivalence without an initial unitality assumption

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Level 7

LEVEL 7

Cross-representation state approximation with a protected finite set

`MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_crossRepresentation_path_approx`

Finite entrywise tests allow new state approximation while an entire unitary path nearly fixes earlier data.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Purifying a vector state on a finite CAR stage, Exact vector transport along an almost central unitary path

USED BY IN THIS PROJECT

Pure CAR states admit two-sided inner intertwining sequences

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LEVEL 7

Approximating another vector state along a unitary path

`MathlibAnnex.CStarAlgebra.CAR.exists_unitary_crossRepresentation_path_approx`

A fixed irreducible CAR representation can approximate any vector state on finitely many elements by moving a given unit vector.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Purifying a vector state on a finite CAR stage, Lifting a corner involution to an ambient CAR unitary, A finite-row average centralizes its matrix stage

USED BY IN THIS PROJECT

Pure CAR states admit two-sided inner intertwining sequences

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LEVEL 7

The cyclic sum fills every irreducible target representation

`MathlibAnnex.CStarAlgebra.CAR.exists_surjective_selectedAtomicCyclicIsometry`

Uses shell reconstruction and residual lines to turn an isometric CAR intertwiner into a unitary equivalence on the source.

IMMEDIATE PREREQUISITES IN THIS PROJECT

A common root vector realizes every selected state through the generators, Assembling selected GNS cyclic subspaces into an isometric source representation

USED BY IN THIS PROJECT

Every irreducible representation is unitarily equivalent to the inclusion

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LEVEL 7

The shell-generated algebra is not an algebra of all compact operators

`MathlibAnnex.CStarAlgebra.CAR.not_isCompactOperatorModel_shellFamilyTarget`

Excludes an injective representation of the infinite-dimensional unital algebra onto all compact operators.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The CAR source map into the fixed shell-family target, The completed CAR algebra is infinite-dimensional

USED BY IN THIS PROJECT

Structural properties of one shell-generated C*-algebra, A Naimark counterexample of continuum norm density

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LEVEL 7

Extending the CAR trace to the fixed shell target

`MathlibAnnex.CStarAlgebra.CAR.exists_state_extension_trace`

The source trace first becomes a state on the existing target algebra.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The CAR source map into the fixed shell-family target, Constructing the normalized trace on the completed CAR algebra

USED BY IN THIS PROJECT

The GNS representation of the target trace extension

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LEVEL 7

Pointed unitary transport between trace-cyclic target representations

`MathlibAnnex.CStarAlgebra.CAR.exists_pointed_unitary_of_trace_of_cyclic`

Source cyclicity and strong shell sums extend the canonical CAR orbit isometry to the whole target.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The CAR source map into the fixed shell-family target, Transported flags vanish on vectors generated by a trace vector, Reconstructing a represented unitary from its shells and residual corner

USED BY IN THIS PROJECT

The unique trace extension is tracial on the whole target, Uniqueness among all state extensions of the CAR trace

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Level 8

LEVEL 8

Pure CAR states admit two-sided inner intertwining sequences

`MathlibAnnex.CStarAlgebra.CAR.hasInnerIntertwiningSequence_of_pure`

Finite-set vector-state transport supplies explicit corrections; their summable conjugation bounds and state estimates yield the required two-sided sequence.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Approximating another vector state along a unitary path, Cross-representation state approximation with a protected finite set, A two-sided inner intertwining sequence for two states

USED BY IN THIS PROJECT

Pure-state homogeneity of the completed CAR algebra

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LEVEL 8

Every irreducible representation is unitarily equivalent to the inclusion

`MathlibAnnex.CStarAlgebra.CAR.ambientInclusion_unitaryEquivalent`

Extends the unitary equivalence from the CAR source to all additional generators and the norm-closed target.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The atomic common range is one embedded GNS line, The cyclic sum fills every irreducible target representation

USED BY IN THIS PROJECT

Closed-ideal simplicity of the shell-generated algebra, Unitary equivalence without an initial unitality assumption

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LEVEL 8

The GNS representation of the target trace extension

`MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation`

The chosen state on the existing target supplies its cyclic GNS representation.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Extending the CAR trace to the fixed shell target

USED BY IN THIS PROJECT

The target represented on the CAR trace GNS space, Faithfulness of the target-state GNS representation, The unique trace extension is tracial on the whole target, Uniqueness among all state extensions of the CAR trace

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Level 9

LEVEL 9

Pure-state homogeneity of the completed CAR algebra

`MathlibAnnex.CStarAlgebra.CAR.homogeneity`

The constructed inner intertwining sequences discharge the supplier hypothesis and give an approximately inner transporting automorphism.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Two-sided inner sequences imply pure-state homogeneity, Pure CAR states admit two-sided inner intertwining sequences

USED BY IN THIS PROJECT

Choosing the CAR shell family from proved homogeneity

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LEVEL 9

Unitary equivalence without an initial unitality assumption

`MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion_unitaryEquivalent_nonUnital`

Shows why allowing nonunital input representations does not enlarge the irreducible representation class of the shell-generated algebra.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The ambient inclusion of the fixed shell-family target, Every irreducible representation is unitarily equivalent to the inclusion

USED BY IN THIS PROJECT

Structural properties of one shell-generated C*-algebra

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LEVEL 9

Closed-ideal simplicity of the shell-generated algebra

`MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_shellFamilyTarget`

Uses a faithful irreducible representation in the unique unitary-equivalence class to rule out proper nonzero closed two-sided ideals.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The ambient inclusion of the fixed shell-family target, Every irreducible representation is unitarily equivalent to the inclusion, The CAR source map into the fixed shell-family target

USED BY IN THIS PROJECT

Structural properties of one shell-generated C*-algebra, Faithfulness of the target-state GNS representation

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LEVEL 9

Uniqueness among all state extensions of the CAR trace

`MathlibAnnex.CStarAlgebra.CAR.existsUnique_state_extension_trace`

Pointed transport identifies arbitrary state extensions before traciality is proved.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The GNS representation of the target trace extension, Pointed unitary transport between trace-cyclic target representations

USED BY IN THIS PROJECT

None in this scope

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LEVEL 9

The unique trace extension is tracial on the whole target

`MathlibAnnex.CStarAlgebra.CAR.traceExtension_mul_comm`

Unitary invariance, strong shell sums, and a closed centralizer extend the trace identity beyond the source.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The GNS representation of the target trace extension, Pointed unitary transport between trace-cyclic target representations

USED BY IN THIS PROJECT

The fixed shell target has a unique tracial state

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LEVEL 9

The target represented on the CAR trace GNS space

`MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation`

A fixed pointed unitary transports the target action onto the Hilbert space constructed from CAR alone.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The GNS representation of the target trace extension, Transported flags vanish on vectors generated by a trace vector, Reconstructing a represented unitary from its shells and residual corner

USED BY IN THIS PROJECT

The fixed algebra on the CAR trace Hilbert space, Faithfulness of the representation on the CAR trace Hilbert space

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Level 10

LEVEL 10

Choosing the CAR shell family from proved homogeneity

`MathlibAnnex.CStarAlgebra.CAR.homogeneityShellFamily`

CAR homogeneity and exact shell matching supply one family whose distinguished component is fixed by an explicit identity choice.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Purity of the completed root state, Exact projection links for one approximately inner automorphism, A fixed representative family of CAR shell data, Pure-state homogeneity of the completed CAR algebra

USED BY IN THIS PROJECT

A simple C^* -algebra with a unique irreducible representation class, The C^* -algebra generated by the CAR representation and the shell unitaries

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LEVEL 10

Structural properties of one shell-generated C^* -algebra

`MathlibAnnex.CStarAlgebra.CAR.shellFamilyEndpoint`

Collects the proved properties of a single generated algebra, keeping the shell-family input separate from its consequences.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The shell-generated algebra is not an algebra of all compact operators, Closed-ideal simplicity of the shell-generated algebra, Unitary equivalence without an initial unitality assumption

USED BY IN THIS PROJECT

A simple C^* -algebra with a unique irreducible representation class

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LEVEL 10

The fixed shell target has a unique tracial state

`MathlibAnnex.CStarAlgebra.CAR.existsUnique_tracial_state`

Every tracial state is forced back to the CAR trace and hence to its unique extension.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Normalization and the trace identity determine the CAR trace, The unique trace extension is tracial on the whole target

USED BY IN THIS PROJECT

None in this scope

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LEVEL 10

Faithfulness of the target-state GNS representation

`MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation_injective`

The ideal dichotomy of the same target forces the constructed nonzero representation to be injective.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The GNS representation of the target trace extension, Closed-ideal simplicity of the shell-generated algebra

USED BY IN THIS PROJECT

Faithfulness of the representation on the CAR trace Hilbert space

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Level 11

LEVEL 11

Faithfulness of the representation on the CAR trace Hilbert space

`MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation_injective`

Unitary conjugation preserves injectivity of the target-state representation.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The target represented on the CAR trace GNS space, Faithfulness of the target-state GNS representation

USED BY IN THIS PROJECT

Faithfulness of the representation on the CAR trace Hilbert space

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LEVEL 11

The C*-algebra generated by the CAR representation and the shell unitaries

`MathlibAnnex.CStarAlgebra.CAR.AtomicCounterexampleAlgebra`

Defines the single algebra T used in the irreducible and separable faithful representations.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Choosing the CAR shell family from proved homogeneity, Realizing a shell family by a faithful irreducible operator algebra

USED BY IN THIS PROJECT

The fixed algebra on the CAR trace Hilbert space, The ambient representation of the fixed algebra

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LEVEL 11

A simple C*-algebra with a unique irreducible representation class

`MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint`

Establishes the properties that make the fixed algebra T a counterexample to Naimark's problem.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Choosing the CAR shell family from proved homogeneity, Structural properties of one shell-generated C*-algebra

USED BY IN THIS PROJECT

No nonzero irreducible representation on a separable Hilbert space

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Level 12

LEVEL 12

The ambient representation of the fixed algebra

`MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleRepresentation`

Defines the inclusion representation $\rho_0 : T \rightarrow B(\mathcal{H})$.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The ambient inclusion of the fixed shell-family target, The C*-algebra generated by the CAR representation and the shell unitaries

USED BY IN THIS PROJECT

No nonzero irreducible representation on a separable Hilbert space

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LEVEL 12

The fixed algebra on the CAR trace Hilbert space

`MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation`

Defines a representation of the same algebra T on the GNS Hilbert space of the CAR trace.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The target represented on the CAR trace GNS space, The C*-algebra generated by the CAR representation and the shell unitaries

USED BY IN THIS PROJECT

Faithfulness of the representation on the CAR trace Hilbert space

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Level 13

LEVEL 13

Faithfulness of the representation on the CAR trace Hilbert space

`MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation_injective`

Proves that the representation $\tilde{\rho} : T \rightarrow B(H_\tau)$ is injective.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The fixed algebra on the CAR trace Hilbert space, Faithfulness of the representation on the CAR trace Hilbert space

USED BY IN THIS PROJECT

Exact norm density of the fixed atomic algebra, A separably represented C*-algebra with no separable irreducible representation

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LEVEL 13

No nonzero irreducible representation on a separable Hilbert space

`MathlibAnnex.CStarAlgebra.CAR.not_isIrreducible_of_separable`

Rules out irreducible representations of T on separable Hilbert spaces by the finite-dimensionality theorem for a single irreducible representation class.

IMMEDIATE PREREQUISITES IN THIS PROJECT

The ambient representation of the fixed algebra, A simple C*-algebra with a unique irreducible representation class

USED BY IN THIS PROJECT

A separably represented C*-algebra with no separable irreducible representation

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Level 14

LEVEL 14

A separably represented C^* -algebra with no separable irreducible representation

`MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint_and_exists_separable_faithful_representation_and_no_separable_irreducible_representation`

The same algebra T has a faithful representation on a separable Hilbert space, but no nonzero irreducible representation on any separable Hilbert space.

IMMEDIATE PREREQUISITES IN THIS PROJECT

No nonzero irreducible representation on a separable Hilbert space, Faithfulness of the representation on the CAR trace Hilbert space

USED BY IN THIS PROJECT

None in this scope

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LEVEL 14

Exact norm density of the fixed atomic algebra

`MathlibAnnex.CStarAlgebra.CAR.hasDensityCharacter_atomicCounterexampleAlgebra`

Combines a cardinality bound from a faithful representation on a separable Hilbert space with a lower bound for every norm-dense subset.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Faithfulness of the representation on the CAR trace Hilbert space

USED BY IN THIS PROJECT

A Naimark counterexample of continuum norm density

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Level 15

LEVEL 15

A Naimark counterexample of continuum norm density

`MathlibAnnex.CStarAlgebra.CAR.existsNaimarkCounterexampleOfDensity_continuum`

Gives a C^* -algebra of norm density $\mathfrak{c}$ whose nonzero irreducible representations form one unitary-equivalence class, without an identification with the compact operators.

IMMEDIATE PREREQUISITES IN THIS PROJECT

Exact norm density of the fixed atomic algebra, The shell-generated algebra is not an algebra of all compact operators

USED BY IN THIS PROJECT

None in this scope

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